American Computer Science League

2025-2026 ● Contest 1: Short Problems ● Elementary Division

1. Computer Number Systems

Convert 7654_8 to base 16.

A. CAF
B. DBE
C. EDB
D. FAC
E. F9C

2. Computer Number Systems

What is the value of 125_{16} in base 10?

A. 69
B. 288
C. 293
D. 297
E. 338

3. Computer Number Systems

Convert 2025_{10} to base 8.

A. 3862
B. 3751
C. 1537
D. 1045
E. 253

4. Computer Number Systems

How many 0s are there in all of the octal numbers when counting from 2025_8 to 2050_8, inclusive?

A. 10
B. 12
C. 15
D. 20
E. 23

5. Computer Number Systems

Blending two colors is done by finding the average of each component. Given the two colors #B058C3 and #9864AE, what is the blended green component? Express this in base 10.

A. 92
B. 93
C. 94
D. 95
E. 96

6. Computer Number Systems

Making a color lighter is done by adding some number to each component. Similarly, making a color darker is done by subtracting some number from each component. What is the color created by subtracting A_{16} from each component of #B9784F?

A. AF6E45
B. AF6845
C. AA7F45
D. A86F45
E. A96E45

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1. 计算机数字系统

将 7654_8 转换为 16 进制。

A. CAF
B. DBE
C. EDB
D. FAC
E. F9C

2. 计算机数字系统

将 125_{16} 转换为 10 进制，**其值是多少**?

A. 69
B. 288
C. 293
D. 297
E. 338

3. 计算机数字系统

将 2025_{10} 转换为 8 进制。

A. 3862
B. 3751
C. 1537
D. 1045
E. 253

4. 计算机数字系统

从**八进制数** 2025_8 到 2050_8（**含**首尾）**之间，共有多少个** 0？

A. 10
B. 12
C. 15
D. 20
E. 23

5. 计算机数字系统

通过计算**两种颜色中**每个分量的平均值来实现**颜色混合，给定两种颜色** #B058C3 和 #9864AE，**混合后绿色的分量是多少？（用** 10 **进制来表示结果）**

A. 92
B. 93
C. 94
D. 95
E. 96

6. 计算机数字系统

通过向每个颜色分量添加一个数值，可以使颜色变浅。类似地，通过从每个颜色分量减去一个数值，可以使颜色变深。那么，从颜色 #B9784F 的每个分量中减去 A_{16} 后，会**生成**什么颜色？

A. AF6E45
B. AF6845
C. AA7F45
D. A86F45
E. A96E45